

AMJAD YAGHI

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EDUCATION

University of British Columbia

Bachelor of Applied Science, Engineering Physics (Co-Op)

Vancouver

Expected May 2027

EXPERIENCE

Generative Models.ai

Toronto

Software Engineer Intern

May 2025 – Aug. 2025

- Designed a **parallelized LLM blog-generation pipeline** that split outlines into per-heading prompts, enforced strict JSON schemas, and recombined validated sections so long-form drafts could be generated quickly and reliably
- Built a **GEO/SEO + readability scorer and website auditor** that analyzes structure, paragraph/sentence lengths, link quality, and headings to produce rewrite recommendations (GEO = optimizing content to be surfaced by LLM search)
- Scraped and analyzed **819 real-world sites**, finding only 12 using llms.txt (robots.txt-style LLM indexing), and wrote a report that directly informed roadmap priority toward llms.txt tooling
- Implemented Django APIs and Temporal workflows to queue long-running analyses and **prevent request collisions**, keeping demos and beta runs stable under concurrent usage

Medical EdTech Startup

Toronto

Full Stack Developer (Contract, Part-time)

Apr. 2025 – Sep. 2025

- Built a voice-first **patient examination simulator** in React + TypeScript (Tailwind) with cross-browser speech-to-text/text-to-speech, live transcript, and auto-send voice detection to mimic timed patient interviews
- Integrated **Gemini for patient responses** and implemented rubric-based semantic scoring with curated acceptable phrases **to reduce hallucinations**, generating per-skill breakdown feedback (communication/diagnosis/history)

STUDENT DESIGN TEAM

UBC Rocket

Vancouver

Software Engineer, Payload Controls

Sep. 2022 – Aug. 2023

- Built Python controls + a PyQt ops GUI for a **microgravity medical experiment**, controlling valves/actuators and logging sensors to test whether blood-clot breakdown methods work in free-fall (**3U+ CubeSat form factor, ~30,000 ft**)

PROJECTS

Autonomous Badminton-Playing Robot (Capstone) 🔗 | C++20, OpenCV, Eigen, CMake, ESP32/STM32

- Designed a modular **C++20 static library** (pimpl, RAII, Eigen) powering a real-time stereo vision pipeline at **~100 FPS** with hardware-synchronized cameras over GPIO triggers
- Triangulated shuttlecock 3D position across a 13m court via stereo rectification and OpenCV's `triangulatePoints`, with an offline ChArUco + court-transform calibration workflow
- Implemented an **Extended Kalman Filter** with hand-derived analytic Jacobian for nonlinear aerodynamic drag, plus RK4 forward simulation to predict landing point within **±5 cm**

Multitasking Simulation Agent 🔗 | Python, ROS, Gazebo, OpenCV, TensorFlow, Qt

- Built a ROS driving agent that reads license plates via OpenCV and a TensorFlow CNN trained on **~1,600** labeled character crops (**>90%** sim accuracy in Gazebo); prototyped imitation learning and shipped PID control for final demo

Autonomous Mario Kart Robot 🔗 | Arduino, STM32 (ARM), C++

- Built a fully autonomous Arduino+STM32 Mario Kart robot using an FSM, **10 kHz** IR-beacon homing (via **convolution filtering**), and sonar edge detection for recovery; built a **zipline claw** that latched reliably to shortcut the course

SKILLS

Languages: Python, TypeScript, JavaScript, C++, C, SQL, MATLAB

Frameworks & Systems: React, Django, ROS, OpenCV, TensorFlow, PostgreSQL, Docker, Temporal, Arduino/STM32/ESP32